



Hospital Engineering and Management

Lecture 5: Nursing Services – Organization, Staffing and Quality of Care

Dr. Sahaj Saxena

Department of Electrical and Instrumentation Engineering
TIET

Course Code: UBM702

Learning Objectives

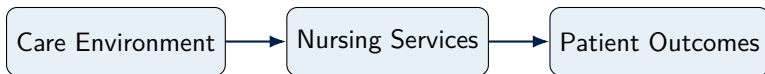
By the end of this lecture, students should be able to:

- ▶ explain the scope and organization of nursing services in a hospital;
- ▶ describe major nursing roles, responsibilities and reporting relationships;
- ▶ identify factors that influence nursing workload, staffing and duty scheduling;
- ▶ relate nursing workflow to patient safety, equipment use and hospital engineering;
- ▶ explain how communication, documentation and handover support continuity of care;
- ▶ identify practical indicators and methods used to improve nursing quality of care.

Connection with the Previous Lecture

Lecture 4 Established

Safe inpatient and intensive-care design depends on appropriate bed-space planning, nursing stations, utilities, medical equipment, infection control and reliable engineering services.



Today's Focus

A well-designed ward becomes effective only when nursing care is organized, staffed, communicated and continuously improved.

Opening Scenario: A Busy Evening Shift

A 30-bed ward receives three new admissions while one patient deteriorates and another requires urgent transfer for imaging. At the same time, an infusion-pump alarm occurs.

Management Questions

- ▶ Who prioritizes the work?
- ▶ Is staffing adequate for the workload?
- ▶ How are tasks delegated safely?
- ▶ What information must be handed over?

Engineering Questions

- ▶ Is equipment immediately available?
- ▶ Are alarms visible and audible?
- ▶ Is backup equipment ready?
- ▶ Can technical support respond quickly?

Nursing service connects patient needs with the hospital system around the clock.

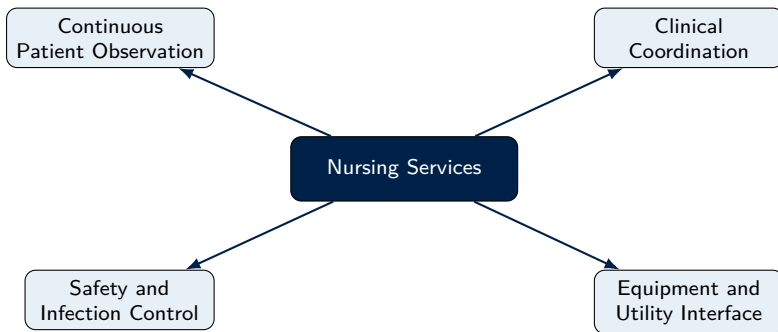
What Are Nursing Services?

Working Definition

Nursing services are the organized activities through which nursing personnel assess patients, plan and deliver care, monitor response, coordinate treatment and maintain continuity and safety.

- ▶ direct bedside care and observation;
- ▶ medication administration and treatment support;
- ▶ patient education and communication;
- ▶ infection-prevention practices;
- ▶ documentation and handover;
- ▶ coordination with doctors, diagnostics and support departments;
- ▶ safe use of medical devices and clinical infrastructure;
- ▶ participation in quality improvement and emergency response.

Why Nursing Services Are Central to Hospital Operation

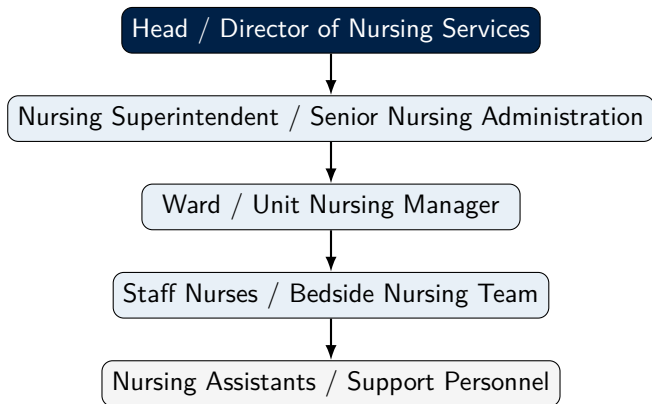


Key Point

Because nurses remain close to the bedside, they often detect deterioration, equipment malfunction and workflow hazards early.

Organization of Nursing Services

A typical hospital uses several levels of nursing responsibility.



Exact titles and reporting structures vary among hospitals; the engineering principle is to create clear accountability and communication channels.

Nursing Service and Nursing Unit: Do Not Confuse Them

Nursing Service

- ▶ hospital-wide professional function;
- ▶ staffing and deployment;
- ▶ policies and standards;
- ▶ training and supervision;
- ▶ quality and safety monitoring.

Nursing Unit

- ▶ physical and operational patient-care area;
- ▶ beds / rooms and nursing station;
- ▶ local staff and equipment;
- ▶ supplies, clean and soiled utility;
- ▶ unit-specific workflows.

Nursing service manages the care system; the nursing unit is one place where that care is delivered.

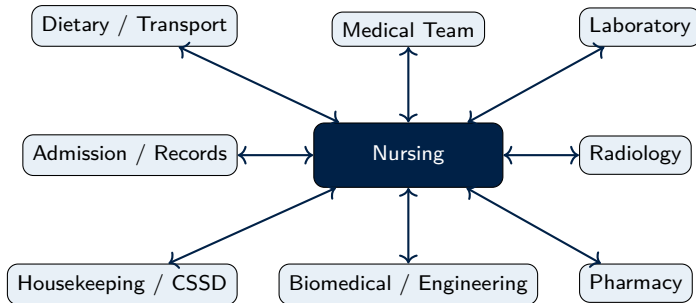
Responsibilities at Different Levels

Level	Typical Responsibilities
Nursing leadership	Policy, manpower planning, budgeting, quality goals, interdepartmental coordination
Unit manager	Shift organization, staff allocation, supplies, local quality indicators, escalation
Staff nurse	Assessment, direct care, medications, monitoring, documentation, communication
Support personnel	Assigned non-professional care and support tasks under appropriate supervision

Management Principle

Authority, responsibility and escalation pathways should be clear before an emergency occurs.

Nursing Services Are Highly Interdependent



Systems View

Delays in diagnostics, pharmacy, equipment repair, transport or housekeeping can appear at the bedside as a nursing-care problem.

What Determines Nursing Staffing?

Staffing should reflect workload rather than bed count alone.

- ▶ number of occupied beds;
- ▶ patient dependency / acuity;
- ▶ admission, transfer and discharge frequency;
- ▶ complexity of procedures;
- ▶ monitoring intensity;
- ▶ isolation requirements.
- ▶ skill mix of available staff;
- ▶ shift length and leave coverage;
- ▶ ward layout and walking distance;
- ▶ documentation workload;
- ▶ technology and automation;
- ▶ local policy and accreditation requirements.

Important

There is no single staffing number suitable for every hospital area or every patient population.

A Simple Workload-Based Staffing Concept

Illustrative Relationship

Daily nursing workload = Average occupied patients $\times$ Care hours per patient-day

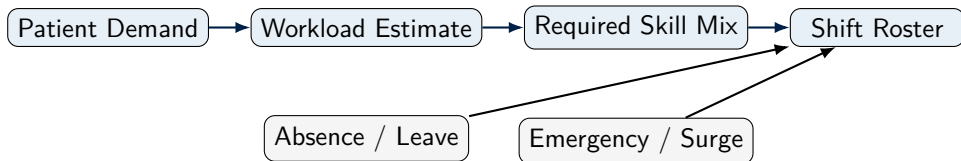
Example for teaching purposes:

- ▶ 20 beds with 85% average occupancy $\Rightarrow$ 17 occupied beds;
- ▶ assumed average nursing requirement = 4 care-hours per patient-day;
- ▶ daily direct-care workload = $17 \times 4 = 68$ nursing-hours;
- ▶ additional allowance is then required for handover, documentation, leave, training and non-direct care.

Do Not Treat This as a Regulatory Ratio

Actual staffing must follow patient acuity, institutional policy, legal requirements and approved professional norms.

From Workload to Duty Roster



Roster Goal

Place the right number of appropriately skilled personnel at the right place and time, while maintaining continuity and reasonable workload.

Why 24-Hour Scheduling Is Difficult

Nursing care continues during nights, weekends and holidays.

Operational Constraints

- ▶ varying patient census;
- ▶ staff leave and absence;
- ▶ uneven skill availability;
- ▶ emergency admissions;
- ▶ overtime and fatigue.

Good Scheduling Tries to Ensure

- ▶ adequate coverage;
- ▶ balanced skill mix;
- ▶ fair rotation;
- ▶ safe handover overlap;
- ▶ contingency capacity.

Patient Acuity and Dependency

Two wards with the same number of occupied beds may require very different staffing.

Patient Type	Typical Care Need	Operational Effect
Stable, ambulatory	routine assessment and treatment	lower direct-care intensity
Partially dependent	assistance, frequent checks	greater bedside time
Highly dependent	extensive support and monitoring	high nursing workload
Critical / unstable	continuous observation, rapid intervention	high skill and infrastructure intensity

Engineering Connection

Higher acuity also increases demand for power outlets, monitoring, medical gases, alarms, pumps and backup systems.

Skill Mix Matters as Much as Head Count

Skill Mix

Skill mix is the combination of personnel with different qualifications, experience and competencies assigned to a clinical area.

A safe team should collectively provide:

- ▶ clinical assessment and escalation capability;
- ▶ medication and procedure competence;
- ▶ device-operation competence;
- ▶ infection-prevention competence;
- ▶ emergency-response capability;
- ▶ supervision of less-experienced staff;
- ▶ communication with patients and multidisciplinary teams.

Ten inexperienced people are not equivalent to ten appropriately skilled people.

Shift Handover: A High-Risk Transition

Handover transfers responsibility and critical information from one caregiver or team to another.

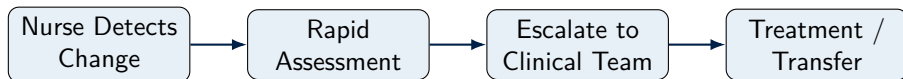
Information to Transfer

- ▶ current condition;
- ▶ treatment and medications;
- ▶ allergies and risks;
- ▶ pending tests / procedures;
- ▶ equipment and lines;
- ▶ required follow-up.

Good Handover Design

- ▶ standardized structure;
- ▶ quiet, interruption-controlled process;
- ▶ read-back / clarification when needed;
- ▶ bedside verification where appropriate;
- ▶ clear documentation of unresolved issues.

Communication Pathway for Patient Deterioration



Failure Mode

A technically correct monitoring system cannot protect the patient if an abnormal value is not recognized, communicated or acted upon.

Documentation and the Electronic Health Record

Nursing documentation creates a time-linked record of patient assessment and care.

- ▶ vital signs and observations;
- ▶ medications and infusions;
- ▶ nursing assessments;
- ▶ intake / output and fluid balance;
- ▶ procedures and responses;
- ▶ education and discharge preparation.

Engineering / IT Considerations

- ▶ reliable network access;
- ▶ bedside terminals or mobile devices;
- ▶ device-data integration;
- ▶ cybersecurity and access control;
- ▶ downtime procedures.

Medication Management: A Systems Process



Risks can arise from:

- ▶ wrong patient, drug, dose, route or timing;
- ▶ look-alike / sound-alike medicines;
- ▶ unreadable or incomplete orders;
- ▶ infusion-pump programming errors;
- ▶ interruptions during preparation;
- ▶ failure to monitor response.

Systems Approach

Safe medication practice depends on people, workflow, storage, labeling, information systems and devices working together.

Nursing Role in Infection Prevention

At the bedside, infection prevention depends on repeated correct actions.

- ▶ hand hygiene at appropriate moments;
- ▶ correct use of personal protective equipment;
- ▶ aseptic technique during procedures;
- ▶ care of invasive lines, catheters and wounds;
- ▶ isolation and transmission-based precautions;
- ▶ safe handling of linen and biomedical waste;
- ▶ environmental cleanliness escalation;
- ▶ surveillance and reporting of suspected infection.

Engineering Link

Handwashing facilities, ventilation, isolation-room pressure, clean water, waste routes and equipment reprocessing all influence nursing infection-control performance.

Patient Safety Responsibilities

Common nursing-sensitive safety risks include:

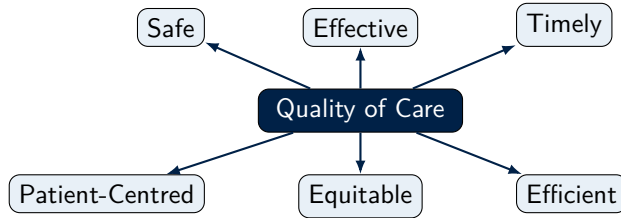
- ▶ patient falls;
- ▶ medication errors;
- ▶ pressure injuries;
- ▶ missed deterioration;
- ▶ identification errors;
- ▶ device-related incidents;
- ▶ line / tube dislodgement;
- ▶ infection risks;
- ▶ communication failures;
- ▶ unsafe transfers.

Design Question

Does the environment make the safe action easier than the unsafe action?

What Does “Quality of Nursing Care” Mean?

Quality should be considered from several dimensions.



Examples of Nursing Quality Indicators

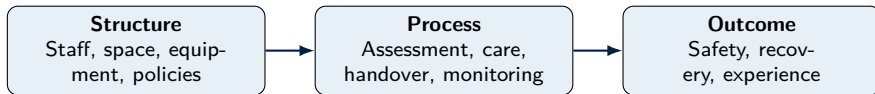
Indicator Area	Possible Measure
Patient safety	falls, medication incidents, pressure injury events
Infection prevention	device-associated infection trends, hand-hygiene compliance
Responsiveness	nurse-call response or escalation time
Care processes	completeness of assessment, medication reconciliation, discharge education
Patient experience	communication, dignity, responsiveness and confidence in care
Workforce	absenteeism, overtime, turnover, training completion
Equipment readiness	device downtime, missing equipment, unresolved safety alerts

Interpretation

A single indicator rarely proves quality. Trends, context and multiple measures should be considered together.

Structure–Process–Outcome Thinking

A useful way to examine quality is to ask three linked questions.



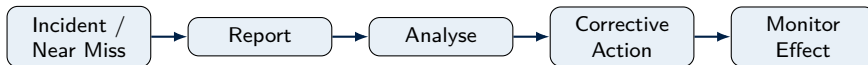
Example

More functioning infusion pumps (structure) can support timely infusion delivery (process), but only when staff are trained and programming errors are controlled; the desired result is safer therapy (outcome).

Incident Reporting and Learning

Purpose of Incident Reporting

The aim is not merely to count errors; it is to identify hazards, contributing conditions and opportunities for prevention.



Safety Culture

Staff are more likely to report hazards when the organization focuses on learning, accountability and system improvement rather than automatic blame.

Root Cause Analysis: Look Beyond the Individual

When an incident occurs, ask what conditions made the error possible.

- ▶ workload and staffing;
- ▶ training and competency;
- ▶ communication;
- ▶ policy and procedure;
- ▶ environment and layout;
- ▶ equipment design;
- ▶ alarm configuration;
- ▶ maintenance status;
- ▶ information-system usability;
- ▶ supervision and escalation.

Example Question

Instead of asking only “Who programmed the pump incorrectly?”, ask “Why was an incorrect setting easy to enter and difficult to detect?”

Nurse Call and Clinical Communication Systems

A nurse-call system is an engineering system with direct influence on patient experience and safety.

- ▶ patient-to-nurse call initiation;
- ▶ visual and audible indication;
- ▶ prioritization of emergency calls;
- ▶ routing to nursing stations or mobile devices;
- ▶ bathroom / toilet emergency calls;
- ▶ event logging and response-time analysis;
- ▶ integration with communication or workflow systems where available.

Reliability Requirement

A call system must be available when the patient is least able to seek help by another route.

Nurses and Biomedical Engineers: A Critical Interface

Nursing Team Provides

- ▶ real-world device-use feedback;
- ▶ incident and alarm observations;
- ▶ cleaning / usability concerns;
- ▶ availability and accessory problems;
- ▶ training needs.

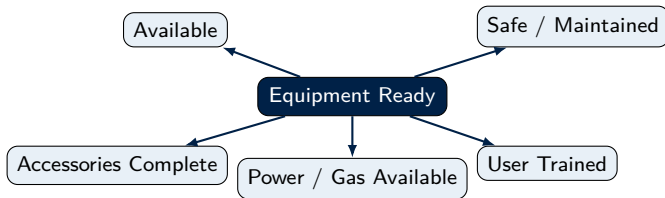
Biomedical Engineering Provides

- ▶ acceptance testing;
- ▶ preventive maintenance;
- ▶ calibration and safety testing;
- ▶ repair and device tracking;
- ▶ technical training and advisories.

Effective equipment management requires continuous clinical–technical communication.

Medical Equipment Readiness at the Bedside

Nurses need more than a functioning device.



Availability vs. Readiness

A ventilator stored in the hospital is not “available for care” if its circuit is missing, battery is depleted, maintenance is overdue or no trained user is present.

Alarm Management and Human Factors

Excessive alarms can reduce attention to meaningful alarms.

- ▶ alarms should indicate a condition that requires awareness or action;
- ▶ alarm limits should match the patient and clinical context;
- ▶ equipment placement should support visibility and audibility;
- ▶ escalation pathways should be defined;
- ▶ nuisance alarms should be investigated rather than routinely ignored;
- ▶ device maintenance and sensor quality influence alarm reliability.

Human-Factors Lesson

A technically loud alarm is not necessarily an effective alarm. It must reach the correct person and communicate useful priority.

Case Study: Infusion Pump Alarm During Peak Workload

A pump repeatedly alarms for occlusion during a busy shift. The patient appears stable and the alarm is silenced several times.

Discuss

1. What immediate nursing checks are required?
2. Could tubing, patient position or pump hardware contribute?
3. When should biomedical engineering be contacted?
4. What should be documented?
5. What system changes could prevent repeated unsafe silencing?

Engineering Lesson

Device incidents often combine clinical workload, alarm design, accessories, maintenance and human behavior.

Case Study: Delayed Transfer to Diagnostic Imaging

A patient scheduled for urgent imaging waits 90 minutes because transport, oxygen cylinder availability and porter communication are not coordinated.

Visible Problem

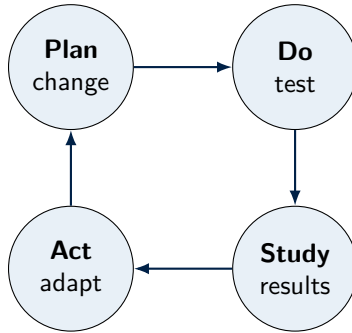
“The patient was transferred late.”

- ▶ patient may deteriorate;
- ▶ diagnostics are delayed;
- ▶ ward staff lose time;
- ▶ imaging schedule is disturbed.

Underlying System Issues

- ▶ transport workflow;
- ▶ equipment availability;
- ▶ communication;
- ▶ ownership of preparation tasks;
- ▶ lack of escalation trigger.

Quality Improvement: The PDSA Cycle



Example

To reduce delayed nurse-call responses, test a new call-routing rule on one ward, measure response time, study unintended effects and then refine the process.

A Simple Nursing Quality Dashboard

Measure	Trend	Possible Action
Patient falls	increasing	review high-risk patients, environment and supervision
Nurse-call delay	increasing	inspect workload, routing and device function
Pump incidents	repeated	device review, training and maintenance analysis
Missed handover items	unstable	standardize handover and reduce interruptions
Equipment downtime	high	improve preventive maintenance / spares planning

Dashboard Principle

Display only measures that lead to interpretation and action. More data do not automatically create better management.

Patient-Centred Nursing Care

Technical efficiency must not remove the human dimension of care.

- ▶ respect patient dignity and privacy;
- ▶ communicate in understandable language;
- ▶ involve patients and families appropriately;
- ▶ respond to pain, comfort and emotional needs;
- ▶ protect confidentiality of health information;
- ▶ consider accessibility and special needs;
- ▶ support safe discharge and self-care education.

Design Connection

Noise, lighting, privacy, toilet access, signage, bedside space and communication technology all influence the patient experience of nursing care.

Fatigue, Workload and Staff Safety

Staff safety and patient safety are interconnected.

Excessive workload can contribute to:

- ▶ delayed tasks;
- ▶ reduced vigilance;
- ▶ communication errors;
- ▶ physical strain;
- ▶ emotional exhaustion.

System controls include:

- ▶ workload-sensitive staffing;
- ▶ safe patient-handling aids;
- ▶ clear escalation routes;
- ▶ reasonable scheduling;
- ▶ supportive team culture.

Engineering Design Can Reduce Nursing Workload

Examples

- ▶ decentralized supplies reduce unnecessary walking;
- ▶ adequate outlets prevent unsafe extension arrangements;
- ▶ equipment tracking reduces search time;
- ▶ well-designed nurse-call routing reduces communication delay;
- ▶ standardized device models simplify training;
- ▶ integrated monitors reduce duplicate data entry;
- ▶ ergonomic bed and lifting equipment reduce staff injury risk.

Good engineering does not replace nurses; it removes avoidable work and hazards.

Classroom Activity: Find the System Failure

Scenario

In one ward, nurses frequently leave the bedside to search for infusion pumps, extension sets and charged batteries. Emergency devices are available somewhere in the hospital, but not always ready when needed.

In groups, identify:

1. the nursing-service problem;
2. the engineering / equipment-management problem;
3. three data items you would collect;
4. one short-term corrective action;
5. one long-term system redesign;
6. one indicator to check whether the intervention works.

Mini Project Ideas

- ▶ nurse-call response-time audit and improvement proposal;
- ▶ equipment-search-time study in a ward or ICU;
- ▶ nursing workload and walking-distance mapping;
- ▶ infusion-pump incident / alarm workflow analysis;
- ▶ bedside equipment-readiness checklist design;
- ▶ handover-quality audit using a structured checklist;
- ▶ ward equipment standardization and training plan;
- ▶ nursing dashboard for safety and engineering indicators.

Expected Output

Define a measurable problem, map the existing process, collect basic observations and propose an engineering-management improvement.

Key Takeaways

- ▶ Nursing services provide continuous coordination of patient care across hospital departments.
- ▶ Staffing depends on patient demand, acuity, skill mix, workflow and local requirements.
- ▶ Handover, documentation and escalation are critical information-management processes.
- ▶ Quality of care should be examined through structure, process and outcome measures.
- ▶ Nursing safety depends on reliable equipment, communication, utilities and hospital engineering support.
- ▶ Incident analysis should search for system causes rather than stopping at individual error.
- ▶ Biomedical engineers and nurses must work together throughout the device life cycle.

Safe nursing care is a product of competent people supported by reliable systems.

Assignment for the Next Lecture

Observe or study one hospital ward, nursing unit or ICU.

Prepare a one-page note containing:

1. a simple nursing-service organization chart;
2. three factors that influence staffing in that area;
3. one important handover or communication process;
4. five frequently used biomedical devices;
5. one equipment-readiness problem that could occur;
6. two possible nursing quality indicators;
7. one engineering-management improvement suggestion.

Submission Format

One page, individually prepared, with one small process diagram or table.

Suggested Reading

- ▶ G. D. Kunders, *Hospitals: Facilities Planning and Management*, Tata McGraw-Hill.
- ▶ B. M. Sakharkar, *Principles of Hospital Administration and Planning*, Jaypee Brothers Medical Publishers.
- ▶ S. Goel et al., *Textbook of Hospital Administration: A Problem-Solving Approach*, Elsevier India.
- ▶ Standard institutional nursing manuals, quality manuals and equipment-management procedures for applied examples.

Next Lecture: Radiology and Imaging Services

Thank You

Questions and Discussion